The Effect of the Bologna Reform on the Graduate Labour Market *

Stanislav Avdeev †

University of Amsterdam and Tinbergen Institute

May 11, 2022

Abstract

In 2011, the Bologna reform shortened the duration of undergraduate programmes by one year and reduced the content of the curricula in Russia. Notably, it did not change the quantity nor the quality of admitted students. Using this reform as a natural experiment, I find no adverse effect of a 20% reduction in the length of higher education on employment. I also find that there are no adverse effects on wages for graduates who spent more time studying, while there are negative effects for students who worked during their studies. I provide suggestive evidence for the compensation–based mechanism to explain these results.

Keywords: difference–in–differences, event–study framework, returns to education, human capital, higher education, Russia

JEL Codes: I23, I26, J24

*I am extremely grateful to Bernardo Pincheira, Victor Rudakov, and Dmitrii Shchetinin for extensive feedback and prolific discussions. I would like to thank Vladimir Gimpelson, Friederike Hertweck, Anna Lukiyanova, Bas Machielsen, Hessel Oosterbeek, Aleksey Oshchepkov, Harry Patrinos, Christopher Walters, colleagues at the Higher School of Economics and the Tinbergen Institute, and three anonymous referees for insightful comments and suggestions. I thank many conference and workshop participants at the 26th Annual Meeting of the Society of Labor Economists, the 35th Annual Congress of the European Economic Association, the 32nd Annual Conference of the Society for the Advancement of Socio–Economic, and the Workshop on Causality in the Social Sciences for their valuable comments. Any errors are those of the author.

†University of Amsterdam and Tinbergen Institute, Amsterdam, the Netherlands; e–mail: s.avdeev@uva.nl

1 Introduction

Investments in education are commonly thought of as productivity–enhancing, and it should be expected that in a competitive labour market education has a positive effect on wages and employment. Despite many papers estimating the causal effects of education on labour market outcomes, the substantial majority of them focuses on high–income countries with developed institutions, competitive labour markets, and high–quality universities (Gunderson and Oreopolous, 2020; Suhonen and Karhunen, 2019; Heckman et al., 2018; Maurin and McNally, 2008; Webbink, 2007). Suhonen and Karhunen (2019) exploit a considerable expansion of the Finnish university system that was gradually rolled out across Finnish regions between the late 1950s and early 1970s as a natural experiment. Using an instrumental variable approach, they find that the annual return to one additional year of higher education is 14% for males and 11% for females. Maurin and McNally (2008) exploit student riots in France in 1968 as a natural experiment and use birth in an affected cohort as an instrumental variable. Due to the events in 1968 and the relaxation of examinations, it became easier to progress to the next stage of higher education and to obtain more years of higher education than would otherwise have been possible. The authors find that each additional year spent at university increases wages by about 14%. Webbink (2007) exploits the institutional reform in the Netherlands in 1982, which reduced the duration of all programmes at research universities from 5 to 4 years, while it did not change the length of programmes at universities of professional education. The author finds that wages for those who graduated from 4–year programmes are substantially lower than wages of graduates completing 5–year programmes. All in all, these studies find systematic positive short– and long–term effects of higher education on labour market outcomes in high–income countries.

Importantly, higher education systems and labour markets in low– and middle–income countries might have distinct and heterogeneous features when compared to high–income countries, such as a more subsidised university sector, lower quality of education, or rigid salaries. Such institutional features influence students' individual decisions to invest money and time to education, and subsequently determine returns to education. To the best of my knowledge, there are only two papers with a credible

identification strategy that estimate the causal effect of higher education on labour market outcomes in low– and middle–income countries. Dai et al. (2021) study a 47% expansion in higher education enrolment between 1998 and 1999 in China. The authors, using a fuzzy regression discontinuity design, find that each additional year of higher education induced by the expansion increases monthly earnings by 24%. Arteaga (2018) exploits a reform at Colombia's top university in 2006, which reduced the amount of coursework and the university study period by half a year but did not change the quality of incoming or graduating students. The author, using a difference–in–differences (DD) design, finds that this reduction led to lower wages and employment, suggesting that human capital plays an important role in the labour market.

This paper estimates the short–term causal effect of one year of higher education on wages and employment in Russia. In Russia, the Bologna reform transformed higher education programmes from 5 to 4 years, while some programmes were not affected by the reform, i.e. remained as 5–year degrees. When undergraduate students entered the labour market together after the reform in 2015, it became possible to estimate the impact of one year of higher education on early labour market outcomes. This reform allows me to use a DD design to estimate the causal effect of one year of higher education (Angrist and Pischke, 2009). The identifying assumption is that, in the absence of the Bologna reform, the quantity and the quality of university applicants and graduates would have evolved similarly in programmes that experienced a one–year reduction of education relative to programmes that did not experience such a change.

The results of the paper show that the OLS estimates of log monthly wages yield returns of about 5% to one year of higher education. These estimates are similar to the results found in the literature based on the Russian labour market. However, my findings suggest that returns to one year of higher education are halved in size and not statistically different from zero. This suggests that the OLS results are biased upwards since people with higher marginal returns to education tend to choose higher levels of schooling. Notably, there are significant adverse effects of the reform on wages of male graduates, while there is no effect for females. I argue that these results are due to different strategies chosen by male and female students. Male students worked more during their studies, while female students

worked less. As a result, female students invested more in the accumulation of their human capital which allowed them to compensate for the reduction in the study duration.

This paper offers several important contributions to the literature. First, it contributes to the emerging literature on the causal estimates of economic returns to higher education in low– and middle–income countries (Dai et al., 2021; Arteaga, 2018). The closest study to mine is Arteaga (2018), who studies a reform at Colombia's top university, but the current paper differs in two important ways. First, Arteaga (2018) studies a reform at one university while I focus on a national reform that affects all universities. Second, Arteaga (2018) estimates an effect of the reform only among graduates in economics and business departments, while the Bologna reform affected students in all fields of study in Russia. Ultimately, my results suggest that a 20% reduction of the university study period does not have a detrimental effect on wages for females, while it affects negatively wages for males. Notably, there are no short–term effects of the reform on employability of graduates.

Second, this paper contributes to the literature focusing on evaluating the results of the Bologna reform in European countries (Neugebauera and Weiss, 2018; Garra, 2013; Farčnik and Domadenik, 2012; Bosio and Leonardi, 2011). This paper is the first to study the impact of the Bologna reform on the labour market outcomes of university graduates in Russia. The core step, in order to implement the reform, is to shorten the length of higher education and create a two–cycle degree structure (bachelor's and master's). The implementation of the reform had mixed effects on the labour market outcomes of graduates in European countries. Although several papers find a significant short–term effect of the Bologna reform (Neugebauera and Weiss, 2018; Farčnik and Domadenik, 2012), other studies find no impact of the reform (Garra, 2013; Bosio and Leonardi, 2011).

Third, this study enriches the literature by suggesting a possible mechanism that explains the heterogeneous effect of shortening the length of higher education on the labour market. Few of the previous papers have attempted to disentangle the mechanisms. The literature aims to set human capital and signalling theories apart, although the theories are not mutually exclusive (Chevalier et al., 2004; Riley, 2001; Groot and Oosterbeek, 1994; Layard and Psacharopoulos, 1974). Arteaga (2018) shows that a reduction in the length of higher education in Columbia resulted in

a reduction of the number of courses which decreased student's performance in recruitment process and ultimately decreased their wages. As a result, Arteaga (2018) rejects a pure signalling model. I also support the human capital theory and show that a possible explanation of the heterogeneous effects is that the reform did not lead to a reduction in the accumulation of human capital of students who could compensate for a one–year reduction by working less during their studies and studying more. To support this claim, I offer suggestive evidence and show that female students worked less during their studies and as a result had to get less on-the-job training due to the insufficient competence.

Finally, I consider three possible threats to the identification strategy, e.g. changes in the cohort composition, an introduction of the standardised exam for admission to universities, and the 2015 Russian financial crisis. First, I show that it is unlikely that university applicants could affect the year of their application as about 90% of school graduates apply at the same year as they graduate from school. Furthermore, I check whether the reform changed the pool of admitted students and graduates. For instance, as earning a higher education degree is easier, lower–ability individuals could be motivated to enrol in 4–year programmes. I show that the reform did not change the quantity nor the quality of admitted students and graduates. Second, I show that the characteristics of students did not change due to an introduction of the standardised exam for admission to universities in 2009. Third, in 2015, there was a financial crisis in Russia, which could have influenced labour market conditions for graduates. To address this concern, I use the event-study framework and include students, who graduated in 2010–2018, to estimate the effect of one year of higher education on employment and find no impact of the reform over longer period of time.

The paper is structured as follows. Section 2 reviews the institutional background of the Russian higher education system and the Bologna reform. Section 3 presents the data and describes the identification strategy. I present the results, robustness checks, and discuss a possible mechanism in Section 4. In Section 5, I present concluding remarks.

2 Institutional background and the reform

The Russian higher education system

All students sit the standardised exam for admission to universities – the Unified State Exam (similar to the SAT) – at their last grade at high school. The higher the score is, the more chances a student gets admitted to a preferred university. Traditionally, about 90% of university applicants are students who apply to university in the same year as they graduate from school.

The Ministry of Science and Higher Education determines the number of state–funded places and the curricula in the majority of universities.[1] The majority of students have state–funded places in higher education institutions in Russia. Some universities, which are under the supervision of sectoral ministries, such as the Ministry of Agriculture, the Ministry of Culture, and the Ministry of Health, set their own regulations for universities, although they have little power to influence the curricula and the quality of programmes.

The Bologna reform in Russia

The two–cycle degree structure has been optional for universities, parallel to 5–year degrees, since 1992. In 2003, Russia signed the Bologna Declaration and some universities started implementing two–cycle degree programmes. In 2011, universities, which were under the supervision of the Ministry of Science and Higher Education and several others, transformed all undergraduate programmes from 5– to 4–year degrees, while other ministries did not require universities to follow this reform.[2] Thus, if applicants applied to a specific programme, they could only apply to either 4– or 5–year programmes. Such an institutional design allows me to use those programmes that were transformed from 5 to 4 years as a treatment group, and those that remained as 5 years as a control group. Importantly, there was no

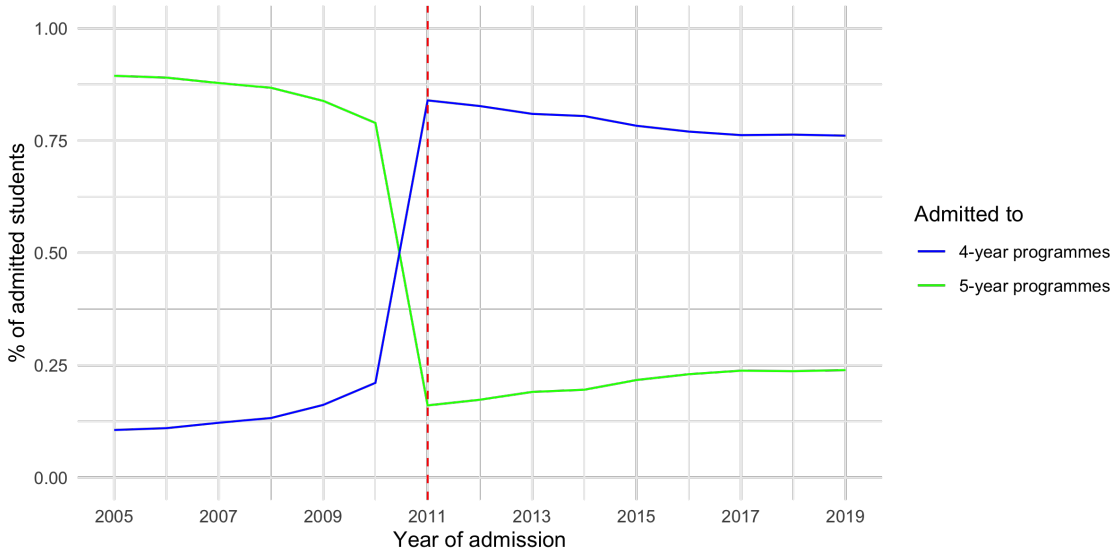
[1] In this paper I describe the system of higher education that is relevant for full–time students in public and private universities. Hereinafter by universities are meant all types of higher education institutions, including universities, institutes, and academies.

[2] The ministries that did not require universities to reduce the length of programmes followed a Decree of the Government of the Russian Federation N1136, which specified certain programmes within broad fields of study to keep at 5 years. It was a politically–oriented decision to keep programmes at 5 years and it did not relate to labour market conditions.

change in the high school curricula to adjust for the reduction in the university study period, so university applicants are homogeneous over time (see Section 4.4 for more details).

The Bologna reform transformed Russian higher education from a unitary system, with 5–year specialists' degrees, to the two–cycle system, with 4–year bachelors' and 2–year masters' programmes. In general, some parts of the course structure and curricula were reduced, especially optional elements. A one–year reduction of the curricula was mostly accompanied by the reduction of specialised courses, so the so–called "specialisation" of higher education was reduced. This is one of the most fundamental institutional reforms of the Russian higher education system in the last decade and one of the most debated ones. The mass introduction of two–cycle higher education degrees happened in the academic year 2010–2011. In 2010, 79% students were admitted to 5–year programmes and 21% to 4–year programmes, and the opposite pattern in 2011: 16% were 5-year students and 84% were 4-year students (see Figure 1). This creates a natural experiment for the 2011 intake.

Figure 1: The share of 4– and 5–year full–time admitted students in public and private universities in 2005–2019



Notes: The red dashed line shows the year of the implementation of the reform.
Source: Ministry of Science and Higher Education.

There are two main reasons that make the case of Russia particularly interesting.

First, the reform was exogenously implemented in the university sector in 2011, thus, universities and students could not affect their decision to participate in the reform or not. I consider the reform as a source of an exogenous variation in the length of the university study period. This reform generates the exogenous assignment of students into the 4–year treatment or 5–year control groups. Since the year of application to university is a function of the year of graduation from school, this policy change provides exogenous variation in the length of the university study period. Second, the duration of programmes is strictly limited to the nominal study period. In other words, students have to graduate within the study limit period. Thus, I assume that any changes in the differences between the treated and control groups can be attributed to the effect of one year of higher education.

Roshchin and Rudakov (2015) and Lukiyanova (2010) summarise a wide variety of papers based on Russian labour market data, which are dedicated to estimating returns to education. All in all, the majority of the studies for Russia relies on OLS estimates. Belskaya et al. (2020) show that returns to one year of higher education are 7.3% in Russia. Kyui (2016) identifies strong positive returns to higher education in terms of employment and wages. Ultimately, there are no causal estimates of how the Bologna reform affected the graduate labour market in Russia, particularly wages and employment.

3 Research design

3.1 Data

The data are obtained from the National Survey of Graduate Employment carried out by the Russian Federal State Statistics Service (Rosstat). This is a repeated cross–sectional sample of students who graduated in 2010–2015. The sample is representative at the national level. The survey provides extensive information on educational backgrounds and labour market experiences during the first years after graduation.

I restrict data to 4– and 5–year university graduates. The education variable refers to the highest level of education attained.[3] According to the definition of

[3]When respondents were asked to specify their higher education degree, they could only choose

youth by Rosstat, I further restrict the dataset to 20–30 year–old graduates who are not currently students. Only graduates from full–time programmes are included. In total, there are 11,332 observations.

I use net monthly wages and net hourly wages, calculated from net monthly wages, the number of working days, and working hours per day.[4] The unemployed category captures both the registered unemployed and the economically inactive.

Table 1 shows the descriptive statistics of the variables for 4– and 5–year university graduates, respectively. 41% and 42% of graduates are males. 94% and 95% of graduates study at public universities. 53% and 60% of students have state–funded places. 82% and 86% of graduates are employed in the year of the survey. The mean monthly wages in 2016 are 23,279 and 24,539 RUB (360 and 380 USD, 1 USD = 65 RUB).

Table 2 shows that 99–100% of 4–year and 92–97% of 5–year students graduate within the nominal study period which is essential for my identification strategy. Importantly, the dropout rate is low so the estimates of returns to education for the analysed sample of graduates would not be biased.

3.2 Identification strategy

This paper estimates the short–term causal effect of one year of higher education on wages and employment in Russia. I do so by using the Bologna reform as a source of an exogenous variation in the length of higher education that is orthogonal to the potential outcomes. Using two cohorts of students who graduated in the same year, I can estimate the differences between the level of wages and the probability of being employed of 4–and 5–year degree holders under similar labour market conditions. However, the simple estimates of returns to higher education for these two groups

between "bachelor's" or "specialist's/master's". Although it may potentially create a bias of the estimates of the wages and employment of the group "specialist's/master's", the share of masters' graduates in 2010–2015 is negligible. The share of masters' graduates is about 4–8% out of all graduates in 2010–2015. Furthermore, I find it most plausible that the inclusion of masters' graduates would bias the estimates towards exaggerating the returns to higher education. Hence, it would most likely overestimate the effect of a one–year reduction of higher education on wages and employment, and such an upward bias would only strengthen the null results found in this study.

[4] A third of employed university graduates did not report their wages. There are no statistically significant differences between university graduates who did and did not report their wages.

Table 1: Summary statistics

Variable	Observations	Mean	SD	Min	Max
4–year university graduates					
Male	1,722	0.41	0.49	0	1
Public institution	1,722	0.94	0.23	0	1
State–funded	1,722	0.53	0.50	0	1
Employed	1,722	0.82	0.38	0	1
Monthly wage, RUB	887	23,279	11,092	5,000	130,000
5–year university graduates					
Male	9,610	0.42	0.49	0	1
Public institution	9,610	0.95	0.21	0	1
State–funded	9,610	0.60	0.49	0	1
Employed	9,610	0.86	0.35	0	1
Monthly wage, RUB	5,320	24,539	12,009	5,000	120,000

Source: National Survey of Graduate Employment carried out by the Russian Federal State Statistics Service.

Table 2: The share of students who did not graduate within the nominal study period and the drop out rate

Variable	2013	2014	2015	2016	2017	2018
% of students who did not graduate within the nominal study period						
4–year programmes	0.00	0.00	0.01	0.01	0.01	0.01
5–year programmes	0.03	0.04	0.05	0.07	0.07	0.07
% of students who dropped out						
4–year programmes	0.11	0.10	0.09	0.09	0.09	0.09
5–year programmes	0.08	0.07	0.07	0.08	0.08	0.08

Source: Ministry of Science and Higher Education.

are confounded by the effect of other changes that had impacts on students enrolled before and after the reform. To correct for that I use a control group, i.e. graduates from programmes that did not change in length. Thus, the changes in the differences between the treated and control groups under certain assumptions are explained by the effect of one year of higher education. A DD design allows me to estimate the average treatment effect on the treated.[5]

The identifying assumption is that, in the absence of the Bologna reform, the quantity and the quality of university applicants and graduates would have evolved similarly in programmes that experienced a one–year reduction of education relative to programmes that did not experience such a change. In other words, I assume that graduates with a particular set of individual characteristics receiving the treatment would have, on average, experienced the same changes in their outcomes had they not received the treatment as individuals with the same set of individual characteristics who did not receive the treatment.

As described in the institutional background, most programmes transformed from 5 to 4 years. The unit of treatment is a programme, i.e. all students follow the nominal study duration of a specific programme. Thus, the treatment group consists of graduates from programmes that changed in length, the control group consists of graduates from programmes that did not change in length. In 2010, almost all programmes lasted for 5 years and the policy change occurred in 2011. As the year of application to university is a function of school graduation year, university applicants could not affect their decision to start 4– or 5–year programmes. In Section 4.4 I show that the Bologna reform indeed did not change the quantity nor the quality of admitted students. As a result, I argue that the Bologna reform offers an exogenous variation in the length of higher education. In other words, variation in the length of the university study period is assumed to be uncorrelated with other factors that might affect labour market outcomes, such as ability or self–selection.

[5] Although a DD design is used to estimate the causal effect of a policy intervention, when some individuals are subject to a treatment and others not, and outcomes are measured in each group before and after the policy reform, individuals in the treated and control groups should not be necessarily the same individuals (Abadie et al., 2017; Athey and Imbens, 2006; Bertrand et al., 2004).

To get naive OLS estimates of returns to education, I use Model 1:

$$Y_{ipry} = \beta_0 + \beta_1 Edu_{ipr} + \mathbf{X_{ipr}} + \lambda_p + \eta_r + \mu_y + \varepsilon_{ipry} \tag{1}$$

where Y_{ipr} is the labour market outcome of graduate i from programme p in region r in year y (I look at log monthly wages and a dummy variable equal to 1 if a graduate is employed and 0 otherwise), where Edu_{ipr} is the imputed number of years of education of graduate i from programme p in region r, $\mathbf{X_{ipr}}$ are exogenous individual characteristics of graduate i from programme p in region r (gender, type of institution (public/private), and funding (state–funded/fee–based)), λ_p are programme fixed effects, η_r are region fixed effects, μ_y are the year of graduation fixed effects.

I conduct a DD analysis to estimate the effect of one year of higher education on labour market outcomes. The results of a DD analysis are obtained from Model 2:

$$Y_{ipry} = \gamma_0 + \gamma_1 Treated_p + \gamma_2 Treated_p * Post_y + \mathbf{X_{ipr}} + \lambda_p + \mu_y + \eta_r + \varepsilon_{ipry} \tag{2}$$

where Y_{ipr} is the labour market outcome of graduate i from programme p in region r, $Treated_p$ is a dummy variable equal to 1 if a student has graduated from a 4–year programme p and 0 otherwise; $Treated_p * Post_y$ is the interaction of $Treated_p$ and a dummy variable for the year of graduation 2015. γ_2 is the parameter of interest that gives the causal estimate of the effect of a one–year reduction of higher education. Only γ_2 is reported in the tables unless stated otherwise. To estimate the probability of being employed, I use a linear probability model. All results reported in this paper are obtained using sample weights. I cluster all standard errors at the programme and the graduation year level.

4 Results

4.1 Main results

In this section I discuss OLS estimates of returns to higher education and present the results of the DD estimates of a one-year reduction of higher education on wages and employment.

OLS estimates of returns to higher education

Table 3 shows the OLS estimates as baseline results of returns to one year of higher education in Russia. Columns (1), (2), and (3) show the association between a one–year of higher education and log monthly wages for all graduates, and separately for males and females. Columns (4), (5), and (6) indicate the results of the linear probability model for employment. The estimated association can be interpreted analogously to returns to one year of higher education. The results for log monthly wages show that returns to one year of higher education are 4.5%. Returns are higher for females than for males, which is consistent with the current literature. The overall results are driven by returns to higher education for females. Returns to one year of higher education are 5.7% for females, while returns to education for males are 3.5%. All the estimates ate statistically significant. The results for employment indicate that a one–year of education is associated with an increase in the probability of being employed by 2.3 pp. The results are driven mostly by female graduates: a one–year of education is associated with 2.7% increase in the employment probability, while the results for males are statistically insignificant. Nonetheless, the OLS results are most likely biased upwards, as endogeneity bias arises due to people with higher marginal returns to education choosing higher level of schooling. Thus, I estimate the causal effect of one year of higher education, using a DD design.

The causal effect of the reform on wages

Table 4 shows the results for log monthly wages and employment, using a DD design. The main variable is *Treated*Post*, which shows the causal effect of a one–year reduction of higher education. Columns (1) and (2) yield the results for log monthly wages. I start with a minimal specification in column (1) that includes only year and region fixed effects. In column (2) I add individual controls, including gender, type of institution (public/private), and funding (state–funded/fee–based). The parsimonious specification in column (1) is the preferred specification. Column (1) shows that returns to one year of higher education are not statistically significant.

The previous findings use log monthly wages to estimate the effect of one year of higher education. One might expect that individuals with more years of education

Table 3: The OLS estimates of the association between years of higher education and log monthly wages and employment

	Log monthly wages			*Employed*		
	All	Male	Female	All	Male	Female
	(1)	(2)	(3)	(4)	(5)	(6)
Years of education	0.045**	0.035*	0.057**	0.023**	0.015	0.027**
	(0.013)	(0.014)	(0.017)	(0.007)	(0.010)	(0.011)
Pre–reform mean	10.02	10.02	10.02	0.88	0.88	0.88
Observations	6,207	2,771	3,436	11,332	4,783	6,549

Notes: Robust standard errors clustered at the programme and the graduation year level in parentheses. The dependent variable is imputed years of education. All columns include year and region fixed effects. Columns (1) and (4) include individual controls for gender, type of institution (public/private), and funding (state–funded/fee–based). Columns (2), (3), (5), and (6) include the same individual controls as columns (1) and (4), excluding gender.

*p<0.1; **p<0.05; ***p<0.01

receive higher monthly wages because they work more hours, or because their productivity increases so their hourly wages increase. Thus, I estimate Model 2, using log hourly wages as a dependent variable. Column (3) of Table 4 yields identical results, suggesting that a one–year reduction does not cause changes in the working hours of graduates. All in all, the results suggest that the OLS estimates of log monthly wages in Table 3 are upward biased and returns to one year of higher education are close to zero and not significantly different from it.

The causal effect of the reform on employment

Similarly, I estimate the effect of a one–year reduction of higher education on employment. Column (5) of Table 4 includes only year and region fixed effects, column (6) adds individual controls, and the parsimonious model is the preferred specification. Column (5) shows that a one–year reduction of higher education does not have the statistically significant effect on employment. Overall, the results confirm that a one–year reduction does not affect the labour market outcomes of university graduates.

Table 4: The effect of a one–year reduction of higher education on log monthly wages, hourly wages, and employment for all

	Log monthly wages		*Log hourly wages*		*Employed*	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated*Post	−0.010	−0.018	−0.021	−0.026	−0.014	−0.015*
	(0.015)	(0.012)	(0.016)	(0.022)	(0.007)	(0.007)
Ind. controls		✓		✓		✓
Pre–reform mean	10.02	10.02	4.91	4.91	0.88	0.88
Observations	6,207	6,207	6,207	6,207	11,332	11,332

Notes: Robust standard errors clustered at the programme and the graduation year level in parentheses. Columns (2), (4), and (6) include individual controls for gender, type of institution (public/private), and funding (state–funded/fee–based).

*p<0.1; **p<0.05; ***p<0.01

4.2 Heterogeneity results

To examine whether the effects of a one–year reduction on wages and employment are heterogeneous across different groups, I estimate the model separately by gender and ability.

By gender

Columns (1) and (2) of Table 8 in the appendix show the results of a reduction of higher education on log monthly wages separately for males and females. I use the parsimonious model and include only year and region fixed effects. There is a negative statistically significant effect of a one–year reduction for male graduates while the effect is positive for female graduates. The estimates for female graduates allow to rule out the effects larger than 3.1%, which is half in size compared to the OLS estimates.

Columns (3) and (4) of Table 8 in the appendix shows the results of a reduction of higher education of employment separately for males and females. The model specification is the same as in columns (1) and (2). In line with previous results for the pooled sample, a one-year reduction does not induce a reduction in the probability of being employed for male and female graduates. Thus, there are no significant gender differences in the effect of a reduction of the length of higher education on employability.

By ability

As a proxy for ability I use a dummy variable for whether a graduate studied at a state–funded or fee–based place. The reason to use this dummy is that applicants with higher scores from the standardised exam for admission are more likely to study at a state-funded place at university, thus, this can be used as a proxy for ability. The results are presented in Table 9. The model specifications are the same as in Table 8. The effect is not statistically significant from zero for wages and employment both for students who study at state–funded places and at fee–based places. This indicates that people with different ability levels do not respond differently to the reform.

4.3 Potential mechanism

Why is a potential explanation for heterogeneous effects of the reform by gender? In this section I provide suggestive evidence about why male and female graduates have been affected differently. I argue that a possible explanation for these results is different strategies chosen by male and female students to compensate for losses in their human capital, i.e. the compensation–based mechanism.

One might argue that 5–year degrees could have been compressed into 4–year programmes without losing much of the human capital because students could study more outside of the classroom. Although I do not have information on the number of hours graduates spend on studying, I can use a proxy for the amount of time students spend on non–studying related activities, for example whether students worked while studying. If 4–year students would have to spend more hours on studying compared to 5–year students, they would have less time to work whilst studying.

Column (1) of Table 5 yields a positive effect of the reform on whether a male student worked while studying, while column (2) shows that there is a negative effect for females. These results provide a potential explanation for why males and females have been affected differently by the reform. Female students decided not to work during their shortened university programme to compensate for human capital losses, while male students chose to work more. Speculatively, male students considered working while their studies as a form of investing in their human capital

in order to compensate for their losses, while female students considered working while studies as a harmful practice for their level of human capital.

To support this claim, one would need to have some proxy for the level of accumulated human capital. Since male students worked more during their studies, while female students worked less, one can expect that their accumulated human capital would differ. As human capital theory suggests, shortening the length of university study period reduces the direct and indirect costs of obtaining a degree (Schultz, 1961; Becker, 1962). The reduction in the length of study leads to the decreased productivity of a graduate, which will result in lower wages and unfavourable labour market opportunities. However, some students can compensate the reduction of the duration of their studies and invest in the accumulation of the human capital by studying more outside of the classroom.

There are some questions in the dataset that help to support this claim. The most relevant question for the paper is whether graduates had to get on-the-job training due to the insufficient competence. About 14% of 4–year and 5–year graduates said that they got on–the–job training due to the insufficient competence. Column (3) of Table 5 shows that there is a positive effect of the reform on the probability of getting on-the-job training for males, while there is a negative effect for female graduates. The estimated coefficients are quite sizable on its own, but especially when they are compared to the pre–reform means.

These results suggest that male graduates decreased their accumulated amount of human capital, while females did not. As a result, male graduates had to take more on-the-job-training than females did. By combining the results for the time spent outside of the classroom and for the accumulated level of human capital, I conclude that female students could compensate for the reduction in the duration of higher education by working less during their studies, such that they did not decrease their level of human capital. Male graduates worked more during their studies, and, as a result, they could not compensate for the losses in their human capital and have been negatively affected by the reform.

Table 5: The effect of a one–year reduction of higher education on combining study and work and the level of on-the-job training

	Combining study and work		*On-the-job training*	
	Male	Female	Male	Female
	(1)	(2)	(3)	(4)
Treated*Post	0.081***	−0.042**	0.072**	−0.045**
	(0.015)	(0.012)	(0.024)	(0.013)
Pre–reform mean	0.28	0.20	0.14	0.13
Observations	4,783	6,549	1,385	1,747

Notes: Robust standard errors clustered at the programme and the graduation year level in parentheses. All columns include year and region fixed effects.

*p<0.1; **p<0.05; ***p<0.01

4.4 Robustness checks

In this section I consider three possible threats to the identification strategy: changes in the cohort composition, an introduction of the standardised exam for admission to universities, and the 2015 Russian financial crisis.

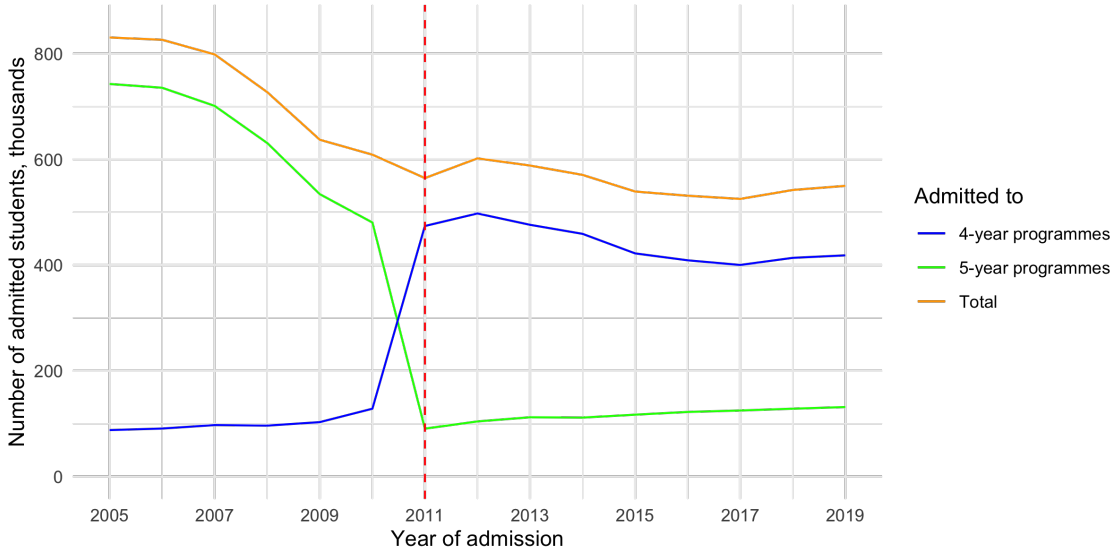
Changes in the cohort composition

One might argue that school graduates could defer the year of application from 2010 to 2011 in order to start a 4–year instead of a 5–year programme. However, 90% of school graduates apply to full–time university programmes in the same year as they graduate from school in the observed period. Thus, there is no evidence for the Bologna anticipation effect as it is unlikely that university applicants could affect the year of their application.

One might expect that the reduction in the length of university study period could attract more students, as earning a degree has reduced the direct and indirect costs. In other words, this reform could have attracted more students into universities, who would not go to university otherwise. Given the shortening of the university study period, lower–ability individuals should be motivated to enrol in these programmes. This might potentially create a bias, if more students were admitted to universities in 2011 and later on, compared to previous years. Figure 2 shows the total inflow of 4– and 5–year full–time students into public and private universities in 2005–2019. The total number of admitted students is decreasing over

time, due to demographic changes. As evident in Figure 2, there is no increased inflow of students in the year of the reform and the following years, suggesting that the reform did not change the pool of admitted students.

Figure 2: The number of 4– and 5–year full–time admitted students in public and private universities in 2005–2019, thousands



Notes: The red dashed line shows the year of the implementation of the reform.
Source: Ministry of Science and Higher Education.

Table 6 shows the descriptive statistics for university students by the year of their graduation. The share of male students, the share of students from public universities, and the share of students who have state–funded places is consistent throughout all years. Overall, the observed characteristics of the cohorts of students are stable over time.

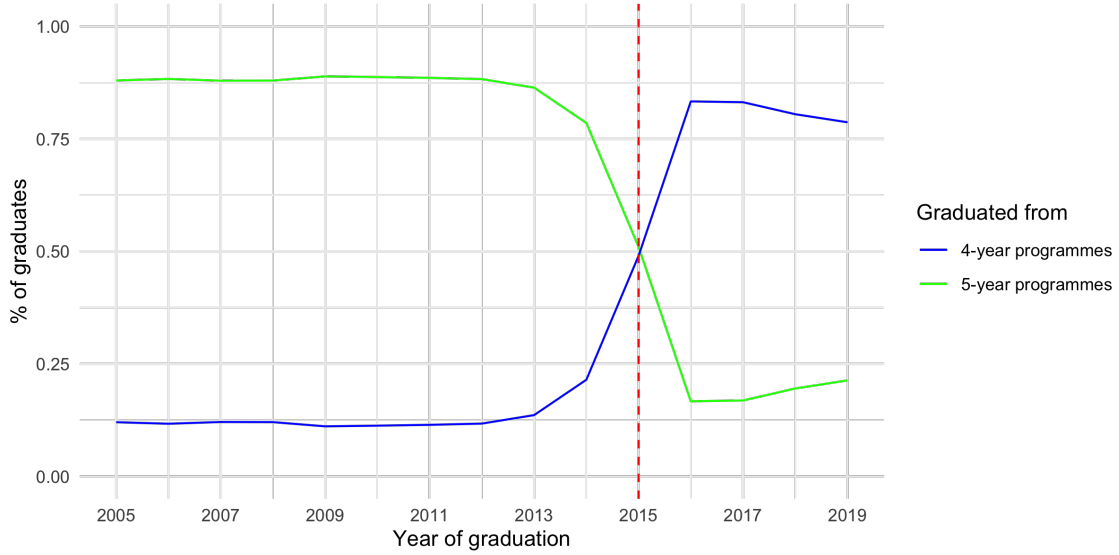
Ultimately, the first cohort of students, who were admitted to 4–year programmes in 2011, graduated in 2015, and the last cohort of students, who were admitted to 5–year degrees in 2010, also graduated in 2015. Thus, the proportion of 4–year graduates to 5–year graduates was 1:1 in 2015 which confirms that there were equal number of university applicants in 2010 and 2011 who could finish their studies on time (see Figure 3). This indicates that the implementation of the reform has not led to a shift in the composition of graduates.

Table 6: The characteristics of university students by the year of graduation

Variable	2010	2011	2012	2013	2014	2015
Male	0.44	0.44	0.45	0.42	0.40	0.39
	(0.50)	(0.50)	(0.50)	(0.49)	(0.49)	(0.49)
Public institution	0.94	0.96	0.95	0.95	0.96	0.95
	(0.23)	(0.20)	(0.22)	(0.22)	(0.19)	(0.21)
State–funded	0.59	0.60	0.59	0.59	0.59	0.58
	(0.49)	(0.49)	(0.49)	(0.49)	(0.49)	(0.49)

Notes: This table shows sample means and standard deviations from the the National Survey of Graduate Employment carried out by the Russian Federal State Statistics Service.

Figure 3: The share of 4– and 5–year full–time graduates in public and private universities in 2005–2019



Notes: The red dashed line shows the year of a double flow of graduates.
Source: Ministry of Science and Higher Education.

An introduction of the standardised exam for admission to universities

The standardised exam for admission to universities – the Unified State Exam (similar to the SAT) – was implemented in some regions in 2003 and was introduced nationwide in 2009. Thus, it increased the accessibility of universities for students with different backgrounds. Table 7 shows the observable characteristics of graduates when they were admitted to university. There are no differences in the age of students at which they were admitted to university: the mean age is 17.6. The share of students from rural areas is slightly higher in the period 2013–2015 than in the period 2010–2012. The share of students from the most developed cities, Moscow and Saint Petersburg, remained stable over the period. Thus, this reform does not influence the results of the paper as students were equally affected by this reform.

Table 7: The characteristics of graduates when they were admitted to university

Variable	2010	2011	2012	2013	2014	2015
Age at the year of application	17.38	17.45	17.62	17.66	17.82	17.73
	(1.09)	(1.21)	(1.26)	(1.27)	(1.39)	(1.35)
Rural residence	0.21	0.23	0.24	0.28	0.27	0.28
	(0.41)	(0.42)	(0.43)	(0.45)	(0.44)	(0.45)
Moscow and Saint Petersburg residence	0.13	0.13	0.13	0.12	0.10	0.12
	(0.34)	(0.33)	(0.34)	(0.32)	(0.30)	(0.33)

Notes: This table shows sample means and standard deviations from the the National Survey of Graduate Employment carried out by the Russian Federal State Statistics Service.

The 2015 Russian financial crisis

Another confounding factor is that there was a crisis in Russia in 2015. The literature shows that students, who graduate in a recession, suffer lower wages and experience unfavourable labour market conditions (Oreopoulos et al., 2012; Kahn, 2010). Russian graduates could have suffered due to the crisis in 2015, which is the year of interest. To check the robustness of my findings, I use data from the same source for students who graduated in 2010–2018.[6] The drawbacks of this dataset

[6]The data are obtained from the Russian Labour Force Survey carried out by the Russian Federal State Statistics Service (Rosstat). The restrictions described in Section **??** are applied to this dataset if possible. More detailed information about this dataset is available upon request.

are that there is no information on whether a student graduated from a part– or full–time programme, and there is no information on wages. Although there might arise a bias, due to the inclusion of part–time graduates, I expect this bias to be consistent for the treatment and control groups. I estimate the following even-study specification:

$$Y_{ipr} = \gamma_0 + \sum_{\tau=-q}^{-1} \gamma_\tau D_{p\tau} + \sum_{\tau=0}^{m} \delta_\tau D_{\tau p} + \mathbf{X_{ipr}} + \lambda_p + \mu_y + \eta_r + \varepsilon_{ipr} \tag{3}$$

where $D_{p\tau}$ is a dummy variable equal to 1 if a student has graduated from a 4–year programme p in year τ and 0 otherwise. I include q leads and m lags, and the reform occurs in year 0.

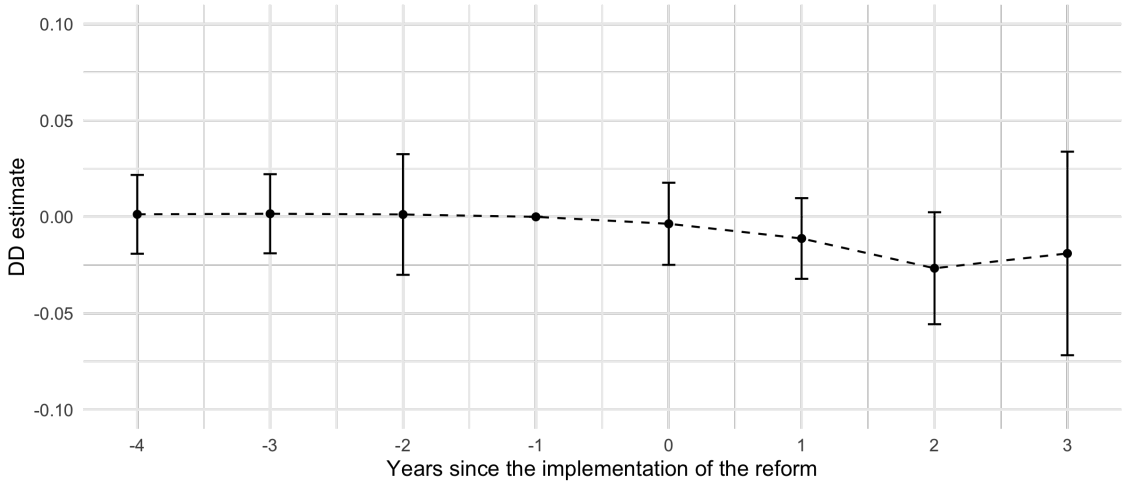
In total there are 4 cohorts of university students who graduated after the reform (2015–2018). The estimates of the effect of the reform on employment in the even-study framework are presented in Figure 4. Although, the magnitude of the effect seems to be increasing over time, there is no evidence to reject the null effect. These results support a zero effect of a one–year reduction of higher education on employment for graduates in 2015–2018.

Figure 4 also provides the estimates of the pre-treatment leads with 95% confidence intervals, i.e. the estimates of the interaction of the treatment dummy and pre–reform period dummies. Ultimately, this test of the parallel pre–trends seems to support the parallel trend assumption, suggesting that prior to 2015 labour market conditions for the treated and control groups followed similar patterns (Cunningham, 2021).

5 Conclusion

The main result of the paper is that it provides more convincing evidence of the causal effect of one year of higher education on labour market outcomes in a middle–income country, i.e. Russia. As I exploit an exogenous variation in the university study period, my findings do not suffer from the endogeneity problem, which the previous literature, based on Russian labour market data, has difficulties addressing. The results show that the DD estimates are smaller than the OLS estimates which

Figure 4: Event-study estimates of a one–year reduction of higher education on employment



Notes: The figure shows the regression coefficients of interaction between the treatment variable and time period dummies with 95% confidence intervals. The omitted period is -1, the year before the implementation of the reform.

is intuitive as a positive relationship between education and unobserved ability is expected which creates an upward bias of the naive OLS estimates.

As forgone earnings and tuition fees are the real costs of going to university instead of working, 4–year graduates have lower costs for earning a higher education degree. This suggests that the reform decreased the university study period, forgone earnings, and tuition fees without negatively affecting labour market outcomes for graduates who chose to work less and study more. The reform has shortened the university study period, allowing 4–year graduates to have an additional year of labour market experience compared to those who completed 5–year programmes. As the literature suggests, additional work experience has an impact on lifetime earnings, meaning the reform has increased the lifetime wealth of 4–year graduates (Dustmann and Meghir, 2005; Topel, 1991).

Ultimately, the results shed light on the causal effects of higher education on the labour market in low– and middle–income countries. A limitation of this paper is that only short–term effects of a one–reduction of higher education are estimated. Further research can use the Bologna reform as a source of an exogenous variation in the length of education and estimate long–term labour market effects. The reform

can be used to study other effects of higher education, such as lifetime wealth, health, or voting, as it offers a rare exogenous variation for producing convincing estimates.

References

Abadie, A., S. Athey, G. Imbens, and J. Wooldridge (2017). When Should You Adjust Standard Errors for Clustering? *NBER Working Paper Series 24003*, 1–28.

Angrist, J. D. and J.-S. Pischke (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton, New Jersey: Princeton University Press.

Arteaga, C. (2018). The Effect of Human Capital on Earnings: Evidence from a Reform at Colombia's Top University. *Journal of Public Economics 157*, 212–225.

Athey, S. and G. W. Imbens (2006). Identification and Inference in Nonlinear Difference-in-Differences Models. *Econometrica 74*(2), 431–497.

Becker, G. S. (1962). Investment in Human Capital: A Theoretical Analysis. *The Journal of Political Economy 70*(5), 9–49.

Belskaya, V., K. S. Peter, and C. M. Posso (2020). Heterogeneity in the Effect of College Expansion Policy on Wages: Evidence from the Russian Labor Market. *Journal of Human Capital 14*(84-121).

Bertrand, M., E. Duflo, and S. Mullainathan (2004). How Much Should We Trust Differences-In-Differences Estimates? *The Quarterly Journal of Economics 119*(1), 249–275.

Bosio, G. and M. Leonardi (2011). The Impact of Bologna Process on the Graduate Labour Market: Demand and Supply. *IZA Discussion Papers 5789*, 1–45.

Chevalier, A., C. Harmon, I. Walker, and Y. Zhu (2004). Does Education Raise Productivity, or Just Reflect It? *The Economic Journal 114*(449), F499–F517.

Cunningham, S. (2021). Difference-in-Differences. In *Causal Inference: The Mixtape*. Yale University Press.

Dai, F., F. Cai, and Y. Zhu (2021). Returns to Higher Education in China – Evidence from the 1999 Higher Education Expansion Using a Fuzzy Regression Discontinuity. *Applied Economics Letters*, 1–6.

Dustmann, C. and C. Meghir (2005). Wages, Experience and Seniority. *The Review of Economic Studies 72*(1), 77–108.

Farčnik, D. and P. Domadenik (2012). Has the Bologna Reform Enhanced the Employability of Graduates? Early Evidence from Slovenia. *International Journal of Manpower 33*(1), 51–75.

Garra, P. (2013). *The Impact of Bologna Process in the Portuguese Labour Market.* Ph. D. thesis, Technical University of Lisbon.

Groot, W. and H. Oosterbeek (1994). Earnings Effects of Different Components of Schooling: Human Capital Versus Screening. *The Review of Economics and Statistics 76*(2), 317–321.

Gunderson, M. and P. Oreopolous (2020). Returns to Education in Developed Countries. In S. Bradley and C. Green (Eds.), *The Economics of Education*, pp. 39–51. Academic Press.

Heckman, J., J. Humphries, and G. Veramendi (2018). Returns to Education: The Causal Effects of Education on Earnings, Health, and Smoking. *The Journal of Political Economy 126*(S1), S197–S246.

Kahn, L. B. (2010). The Long-Term Labor Market Consequences of Graduating from College in a Bad Economy. *Labour Economics 17*(2), 303–316.

Kyui, N. (2016). Expansion of Higher Education, Employment and Wages: Evidence from the Russian Transition. *Labour Economics 39*, 68–87.

Layard, R. and G. Psacharopoulos (1974). The Screening Hypothesis and the Returns to Education. *The Journal of Political Economy 82*(5), 985–998.

Lukiyanova, A. (2010). Returns to Education in Russia: Evidence from Meta-Analysis [Otdacha ot obrazovaniya: chto pokazyvaet meta-analiz]. *HSE Economic Journal 14*(3), 326–348.

Maurin, E. and S. McNally (2008). Vive la Révolution! Long-Term Educational Returns of 1968 to the Angry Students. *Journal of Labor Economics 26*(1), 1–33.

Neugebauera, M. and F. Weiss (2018). A Transition without Tradition: Earnings and Unemployment Risks of Academic versus Vocational Education after the Bologna Process. *Zeitschrift für Soziologie 47*(5), 349–363.

Oreopoulos, P., T. von Wachter, and A. Heisz (2012). The Short- and Long-Term Career Effects of Graduating in a Recession. *American Economic Journal: Applied Economics 4*(1), 1–29.

Riley, J. G. (2001). Silver Signals: Twenty-Five Years of Screening and Signaling. *Journal of Economic Literature 39*(2), 432–478.

Roshchin, S. and V. Rudakov (2015). Do Graduates Wages Measure the Quality of Education? A Review of Existing Studies. *Educational Studies 1*, 138–171.

Schultz, T. (1961). Investment in Human Capital. *The American Economic Review 51*(1), 1–17.

Suhonen, T. and H. Karhunen (2019). The Intergenerational Effects of Parental Higher Education: Evidence from Changes in University Accessibility. *Journal of Public Economics 176*, 195–217.

Topel, R. (1991). Specific Capital, Mobility, and Wages: Wages Rise with Job Seniority. *The Journal of Political Economy 99*(1), 145–176.

Webbink, D. (2007). Returns to University Education: Evidence from a Dutch Institutional Reform. *Economica 74*(293), 113–134.

6 Appendix

Table 8: The effect of a one–year reduction of higher education on log monthly wages and employment separately by gender

	Log monthly wages		*Employed*	
	Male	Female	Male	Female
	(1)	(2)	(3)	(4)
Treated*Post	−0.046**	0.017	−0.020	−0.012
	(0.015)	(0.015)	(0.032)	(0.022)
Pre–reform mean	10.13	9.92	0.95	0.82
Observations	2,771	3,436	4,783	6,549

Notes: Robust standard errors clustered at the programme and the graduation year level in parentheses. All columns include year and region fixed effects.

*p<0.1; **p<0.05; ***p<0.01

Table 9: The effect of a one–year reduction of higher education on log monthly wages and employment separately by funding

	Log monthly wages		*Employed*	
	State–funded	Fee–based	State–funded	Fee–based
	(1)	(2)	(3)	(4)
Treated*Post	−0.013	−0.001	−0.001	−0.021
	(0.020)	(0.010)	(0.015)	(0.015)
Pre–reform mean	10.02	10.01	0.88	0.86
Observations	3,730	2,477	6,697	4,635

Notes: Robust standard errors clustered at the programme and the graduation year level in parentheses. All columns include year and region fixed effects.

*p<0.1; **p<0.05; ***p<0.01